

Md. Tanvirul Islam

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Flat: H-08, EHL Nabarun Residences, Section: 2, Mirpur, Dhaka - 1216, Bangladesh

OBJECTIVE

Graduated in Electrical and Electronic Engineering from the Bangladesh University of Engineering and Technology (BUET), with research interests spanning power system optimization, energy economics, and the low-carbon energy transition. Experienced in the modeling, optimization and analytical evaluation of modern power systems, with particular emphasis on carbon-aware system operation, security-constrained optimization and the integration of advanced generation technologies. Motivated by the development of computational methodologies that support resilient, economically efficient and sustainable power system planning and operation. Additionally, interested in nuclear energy systems and other advanced low-carbon technologies as integral components of future decarbonized energy infrastructures.

EXPERIENCE

- **Q-PACERS (DSMC) Research Group**  April 2025 - June 2026
Undergraduate Research Member Dept. of EEE, BUET, Dhaka
 - Conducting simulation-based analysis of security-constrained, carbon-aware optimal power flow (OPF) for low-carbon grid operation.
 - Investigating the integration of advanced generation technologies, including supercritical CO_2 cycles, within OPF frameworks to assess their impact on system efficiency and stability.
 - Developing optimization models incorporating operational constraints and emission signals to support reliable and sustainable power system operation.

EDUCATION

- **Bangladesh University of Engineering and Technology (BUET)** January 2022 - June 2026
B.Sc. in Electrical and Electronic Engineering Dhaka, Bangladesh
 - Major: Electrical Engineering and Power Systems
 - CGPA: 3.51/4.00
- **Govt. Ananda Mohan College** 2020
Higher-Secondary School Certificate (HSC) Mymensingh, Bangladesh
 - GPA: 5.00/5.00
- **Mymensingh Zilla School** 2018
Secondary School Certificate (SSC) Mymensingh, Dhaka
 - GPA: 5.00/5.00

CORE COURSES

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|--|---------------------------------|---------------------------------------|--|
| • Power System Operation and Control | • Engineering | • Communication Systems (2 Courses) | • Electrical Service Design |
| • Power System Protection | • Smart Grid | • Control Systems | • Numerical Techniques |
| • Power System Transmission and Distribution | • Nuclear Engineering | • Digital Signal Processing | • Probability and Statistics |
| • High Voltage | • Energy Conversion (2 Courses) | • Microprocessor and Embedded Systems | • Linear Algebra |
| | • Digital Electronics | | • Electrical and Electronic Circuits (4 Courses) |

RESEARCH INTERESTS

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|---|---|
| • Power System Operation and Planning | • Low-Carbon Energy Systems |
| • Power System Optimization | • Electricity Markets and Energy Economics |
| • Distributed Energy Resources and Grid Modernization | • Data-Driven Methods for Power System Analysis |
| • Renewable Energy Integration | |

PROJECTS

- **FIDVR Mitigation in Power System using Grid-Connected PV Generators and SVCs** 2025
Tools: MATLAB, PSAP
 - Modeled and simulated the impact of grid-connected PV generators and SVCs on mitigating FIDVR events in power systems.
 - Analyzed voltage recovery improvements and developed control approaches to enhance system stability during fault conditions.
- **Smart Curtain Mechanism for Adaptive Directional Sunlight and Intensity Control** 2025
Tools: Arduino IDE
 - Developed a smart curtain system that dynamically adjusts to control both the direction and intensity of sunlight entering indoor spaces.
 - Designed and implemented adaptive algorithms to optimize natural lighting for enhanced energy efficiency and occupant comfort.
 - Integrated sensor feedback to enable real-time curtain positioning for effective sunlight management throughout the day.
- **Determination of Bus Fault Current and Breaker Ratings for LG and LLG Faults** 2024
Tools: MATLAB, PSAP
 - Developed a detailed MATLAB-based power system model to simulate **LG and LLG faults**, analyzing fault currents across critical buses under various network conditions.
 - Determined optimal **circuit breaker ratings** based on simulated fault current magnitudes, ensuring alignment with industry safety standards and operational reliability.
 - Provided practical, data-driven recommendations for **protective device selection**, contributing to improved fault management strategies in modern power system engineering.
- **Meat (Beef) Freshness Detection using E-Nose System** 2024
Tools: MATLAB, Python
 - Designed and implemented an **Electronic Nose (E-Nose) system** using a gas sensor array to detect volatile organic compounds (VOCs) associated with beef spoilage.
 - Applied **machine learning algorithms** to analyze sensor data, enabling accurate classification of beef samples into fresh, moderately aged, and spoiled categories.
 - Demonstrated a practical, low-cost solution for **real-time freshness detection**, contributing to improved quality control in the meat industry and broader food monitoring applications.

INDUSTRIAL EXPERIENCE

- **Bangladesh Power Development Board** November 2025
Industrial Attachment Shahjibazar Power Plant, Habiganj, Bangladesh
 - Studied the operation of a combined-cycle power plant consisting of gas turbine, heat recovery steam generator (HRSG) and steam turbine units.
 - Observed gas turbine operation, including combustion process, generator synchronization, auxiliary systems and operational monitoring procedures.
 - Analyzed power generation workflow, plant instrumentation and supervisory control systems used for real-time monitoring and operation.
 - Studied electrical systems including generator protection, switchgear, transformers, excitation systems and synchronization mechanisms.
 - Gained practical exposure to combined-cycle efficiency improvement, maintenance practices, and operational constraints affecting power plant reliability.
 - Examined the role of thermal power generation in grid support, including load dispatch, frequency regulation, and integration with the national power system.

PATENTS AND PUBLICATIONS

- [C.1] Md. Mostafizur Rahman, Md. Tanvirul Islam, Rony Ghosh, Nahian Tasnim. **Smart Curtain System for Real-Time Adaptive Directional Sunlight Control Using ESP-NOW**. In *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, pp. 382-387. 2025, Dhaka, Bangladesh. DOI: 10.1109/RAAICON69033.2025.11502186
- [C.2] Md. Tanvirul Islam. **“Cross-Scale Performance Analysis of Metaheuristic Algorithms for Simultaneous DG and DSTATCOM Placement in Radial Distribution Networks,”** accepted in *IEEE Region 10 Conference (TENCON)*, Bali, Indonesia, 2026.
- [C.3] Md. Tanvirul Islam. **“Hybrid Analytical-Machine Learning Framework for Ripple Factor Estimation in Cockcroft-Walton Voltage Multipliers with Residual Correction for Non-Ideal Effects,”** accepted in *International Conference on Electrical, Computer and Energy Technologies (ICECET)*, Rome, Italy, 2026.

SKILLS

- **Programming Languages:** Python, C, C++, MATLAB, Verilog, Assembly Language
- **Power System and Simulation:** CYME PSAF, Siemens PSSE, SPICE Circuit Simulator, DIGSilent Powerfactory, Proteus
- **Data Science & Machine Learning:** Python (Pandas, Numpy, scikit-learn, PYOMO), Tensorflow (Basic), R(Basic)
- **Hardware and Digital System:** Verilog (FPGA), STM Microcontroller, Embedded System, Arduino, DeLorenzo Relays, LabVIEW
- **Mathematical & Statistical Tools:** MATLAB and Simulink, Python, R (Basic), Excel (Advanced)

HONORS AND AWARDS

- **Best Leo of the Year** 2020-21
Lions Club International, District 315A3, Bangladesh
 - Awarded for outstanding leadership, community service and dedication to the values of the Leo Club program
 - Recognized for exceptional commitment to social initiatives, teamwork and inspiring positive impact among peers and the wider community
- **Best Student of the Year** 2017
Mymensingh Zilla School
 - Recognized for outstanding academic performance combined with active participation in extracurricular activities, including competitive debating and organizing cultural programs.
 - Demonstrated strong leadership, communication and event management skills
- **Finalist** 2019
Tiger Challenge, MIT Solve
 - Selected as one of the 15 finalists out of 800+ teams in the Bangladesh
 - Recognized as the youngest team among the finalists—highlighting our innovation and entrepreneurial spirit at an early stage.

LEADERSHIP EXPERIENCE

- **Vice President** July 2022 - June 2023
Leo Club International, District 315A3, Bangladesh
 - Led and coordinated multiple community service initiatives, promoting youth leadership and civic engagement
 - Supervised a team of volunteers, ensuring smooth execution of charity drives, blood donation camps, and awareness campaigns throughout the country
 - Facilitated collaboration between clubs and district leadership, enhancing program impact and member participation
- **Treasurer** July 2024 - June 2025
IEEE PES BUET Student Branch Chapter
 - Managed the chapter's financial planning, budgeting, and expense tracking with transparency and accuracy
 - Secured and allocated funds for technical seminars, workshops, and IEEE-sponsored events
 - Collaborated with executive members to ensure financial sustainability and compliance with IEEE financial policies

VOLUNTEER EXPERIENCE

- **Campus Ambassador** 2023
PTAK Prize Case Competetion and Scholarship Program 2023, ISCEA Bangladesh
- **Advisor** 2018 - 2023
Tarunno (A Social Organization at Mymensingh)

PROFESSIONAL MEMBERSHIPS

- **Student Member, IEEE**, Membership ID: 98585506 July 2022 - Present

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Fluent)

Interests: Youth Leadership and Community Volunteering, Entrepreneurship and Social Innovation, Exploring History and Geopolitics, Writing and Creative Expression

REFERENCES

1. Dr. Celia Shahnaz

Professor, Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Email: celia@eee.buet.ac.bd
Relationship: Academic Advisor

2. Dr. Sajid Muhaimin Choudhury

Professor, Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Email: sajid@eee.buet.ac.bd
Relationship: Thesis Supervisor